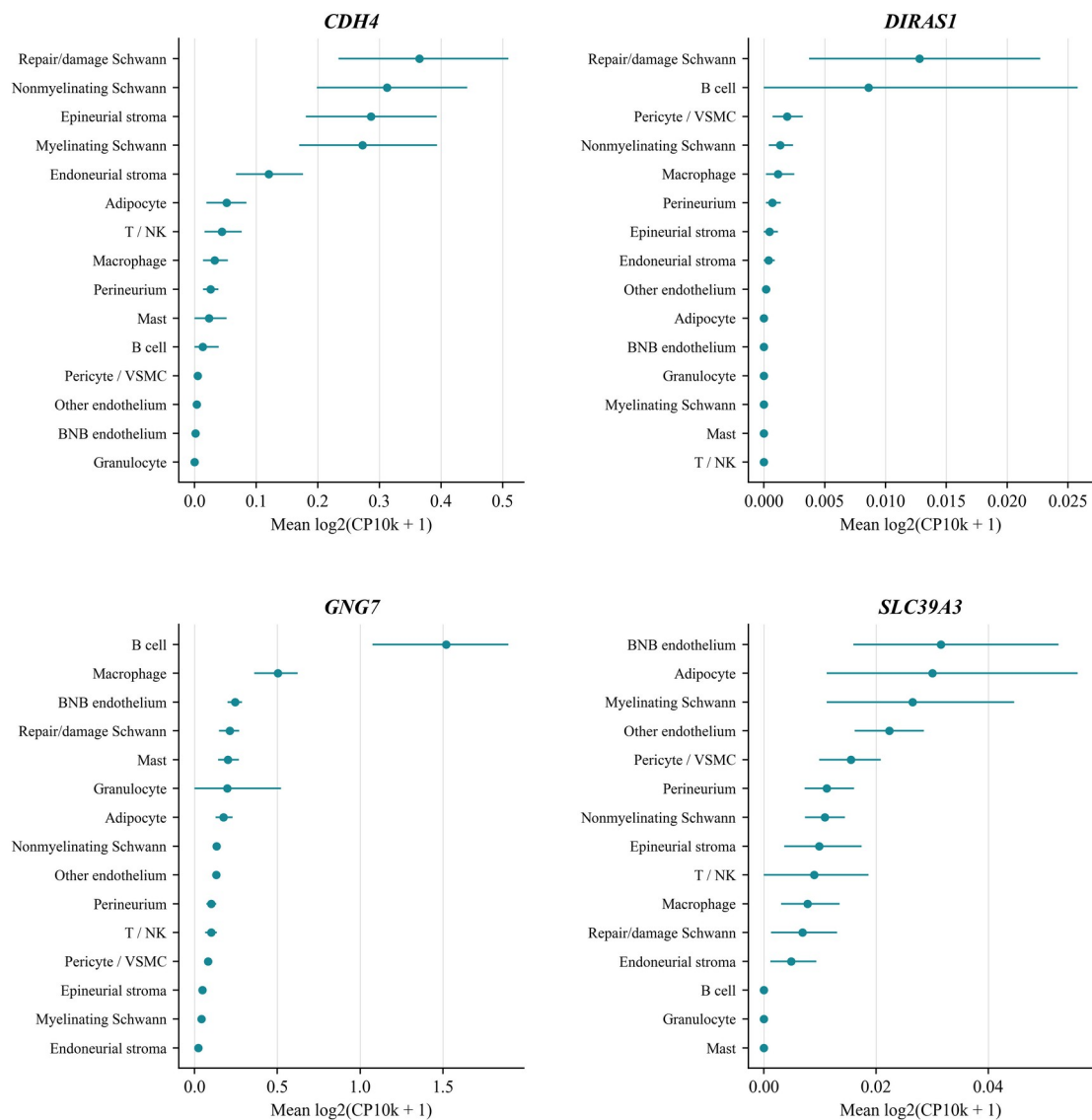


1 **Additional file 3. Supplementary figures**

2 Immune gene and protein profiles across blood, cerebrospinal fluid and peripheral nerve in

3 inflammatory neuropathies

4 **Supplementary Figure 1. Donor bootstrap intervals for genetic candidate expression**



5

6 Points show mean log2(CP10k + 1) across eligible CIDP donors. Bars show percentile 95%

7 intervals from 5000 resamples within each gene and cell group. Resampling was performed at

8 donor level with random seed 20260904. The intervals quantify uncertainty in the donor mean

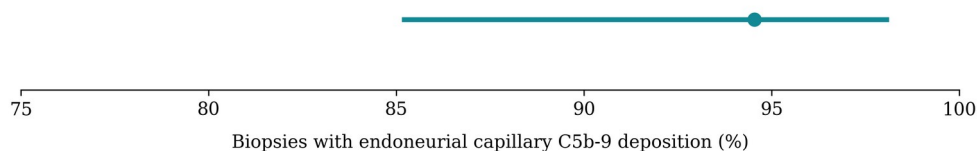
9 for each gene and cell group. The source values and eligible donor counts are provided in the

10 Genetic_bootstrap worksheet of Additional file 1.

12 **Supplementary Figure 2. Published nerve observations and complement deposition**

Published CIDP nerve observations

52 of 55 biopsies 94.5% (Wilson 95% CI 85.1–98.1%)



Additional source observations

Proteomic subset 9 CIDP cases and 2 controls

Higher C3, C5 and C9 abundance was reported.

C6 protein was elevated in 2 of 9 cases.

Histology identified CD68 positive macrophages and CD8 positive T cells.

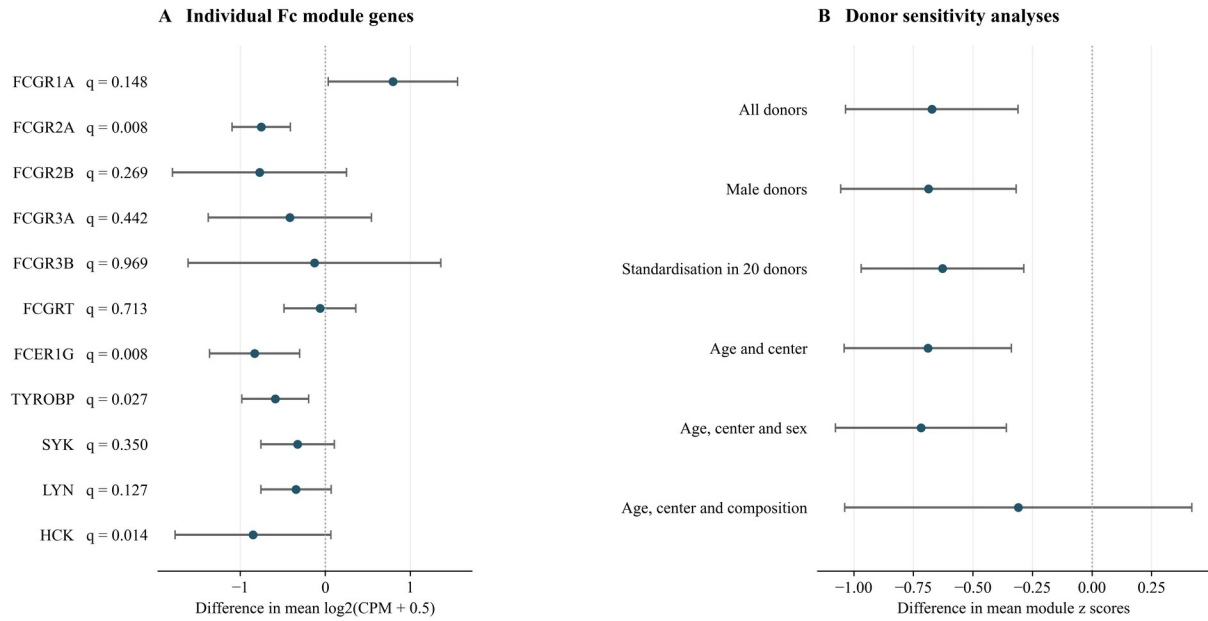
Stascheit et al. 2025 <https://doi.org/10.1007/s00401-025-02936-w>

13

14 Stascheit et al. reported deposition of the terminal complement complex on endoneurial
15 capillaries in 52 of 55 biopsies from patients with CIDP or its variants, together with
16 histological and proteomic findings [37]. The proportion of 94.5% and Wilson 95% interval
17 of 85.1% to 98.1% were calculated from the published count. The other observations
18 summarise qualitative findings from the source study associated with PXD056286. The
19 proteomic subset comprised nine CIDP cases and two controls. Higher C3, C5 and C9
20 abundance was reported, while C6 protein was elevated in only two cases.

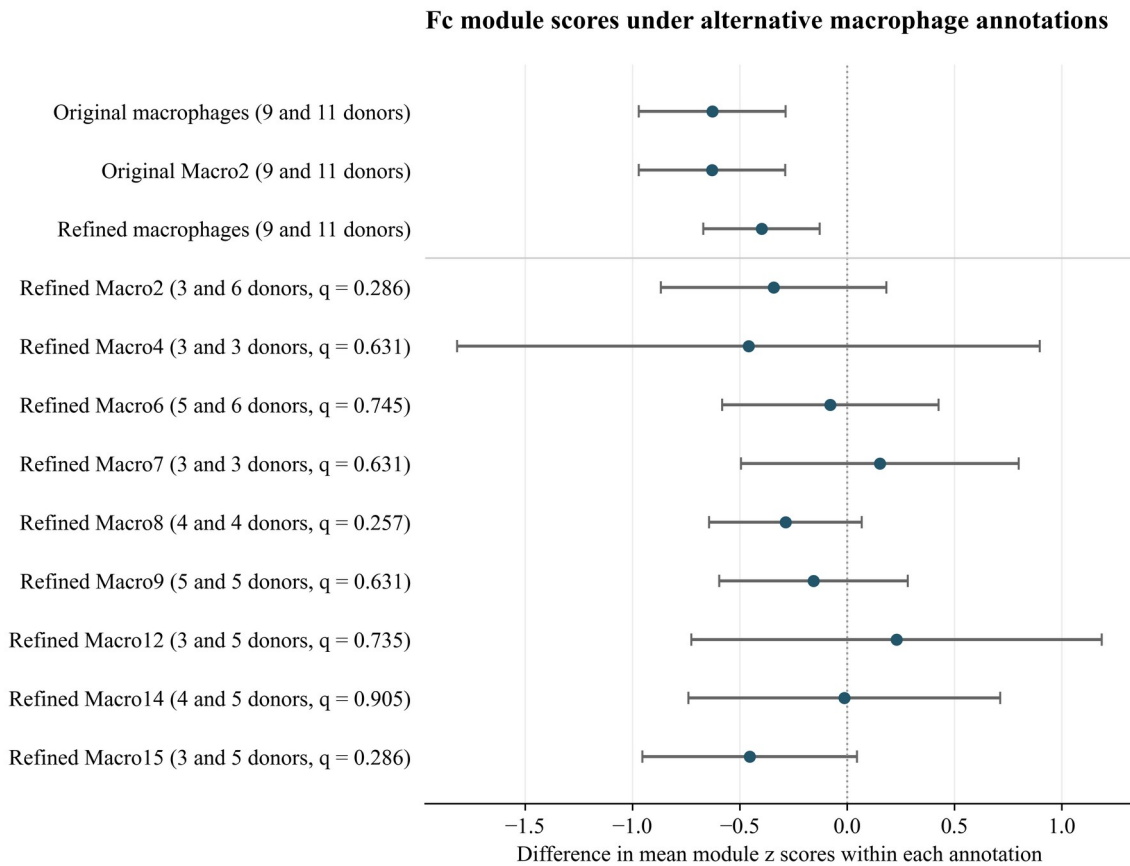
21

Supplementary Figure 3. Fc module components and donor sensitivity analyses



(A) Differences in mean $\log_2(\text{CPM} + 0.5)$ for the 11 measured genes, calculated as CIDP minus CIAP. *FCGR2C* was absent from the matrix. Displayed q values were derived from two sided Mann and Whitney tests with Benjamini and Hochberg correction across the 11 genes. Points and bars show mean differences and unadjusted 95% confidence intervals from separate ordinary least squares models with HC3 covariance and residual degrees of freedom. These intervals do not correspond to the rank tests used to calculate q values. (B) Module score differences under demographic and composition sensitivity analyses, with HC3 95% confidence intervals. Standardisation within the 20 CIDP and CIAP donors changes the score scale. The exact permutation test within centres gave $P = 0.0048$ and is reported separately from the model intervals. The original module q value was 0.0087 within eight macrophage modules and 0.087 across all 80 eligible module tests. These exploratory analyses followed the initial Fc module observation.

Supplementary Figure 4. Fc scores under alternative macrophage annotations



The original and refined annotations represent distinct classification levels. The refined union contains Macro1 through Macro18. Eligibility requires at least 20 nuclei per donor and three donors per diagnosis. Original Macro1 was ineligible. Points and bars show mean score differences calculated as CIDP minus CIAP, with unadjusted HC3 95% confidence intervals. Genes are standardised across eligible CIDP and CIAP donors within each annotation. Parentheses give CIDP and CIAP donor counts. Subtype q values were derived from two sided Mann and Whitney tests with Benjamini and Hochberg correction across nine subtypes, and all exceed 0.05. The three broader definition checks form a separate testing family. The HC3 intervals describe mean differences and do not correspond to the rank tests used for q values.